CO3 - Assessment Tool 1: Case Study Answers

Case Study 1: Smart Medical Diagnosis System

(a) Symbols: F=Fever, C=Cough, R=Rash, B=Breathlessness, Fl=Possible Flu, M=Possible Measles, Pn=Possible Pneumonia.
Rules: R1: F□C→Fl | R2: F□R→M | R3: C□B→Pn.
Facts: F=True, C=True, B=True.

(b) Forward Chaining: Facts F,C,B true. Fire R1 (F□C→Fl): derive Fl (Possible Flu). Fire R3 (C□B→Pn): derive Pn (Possible Pneumonia). R2 cannot fire (R unknown). Result: Possible Flu, Possible Pneumonia.

(c) Backward Chaining goal Pn: Pn needs C□B (R3). C=True (fact), B=True (fact) → both satisfied → Pn confirmed.
Goal tree: Pn → [C (fact-True), B (fact-True)] → Goal proven.

Case Study 2: Autonomous Traffic Management Agent

(a) Unify Vehicle(x) with Vehicle(Ambulance): θ={x/Ambulance}. EmergencyType(x) consistent with θ. MGU θ={x/Ambulance}. Result: ClearPath(Ambulance).

(b) CNF clauses: 1. ¬Vehicle(x)□¬EmergencyType(x)□ClearPath(x); 2. ¬ClearPath(y)□GreenSignal(y); 3. ¬ClearPath(y)□¬RedSignal(y); 4. ¬Vehicle(x)□¬Behind(x,y)□¬GreenSignal(y)□Proceed(x); Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA,Ambulance). Negated goal: ¬Proceed(CarA).
Steps: Resolve 1+facts{x/Ambulance} → ClearPath(Ambulance). Resolve with 2{y/Ambulance} → GreenSignal(Ambulance). Resolve 4 with facts{x/CarA,y/Ambulance} and GreenSignal(Ambulance) → Proceed(CarA). Resolve with negated goal → empty clause □. Proceed(CarA) proven.

(c) Not sufficient - KB has no conflict-resolution rule when two emergency vehicles need GreenSignal simultaneously at the same junction. Need added predicates such as Priority(x), arrival-time ordering, and a rule preventing simultaneous conflicting GreenSignal assignments.

Case Study 3: Agricultural Expert Reasoning System

(a) P1: ¬SoilDry□IrrigationNeeded. P2: ¬IrrigationNeeded□¬CropWheat□ApplyDripMethod. P3: ApplyDripMethod□¬CropWheat□CropAtRisk. Facts (unit clauses): SoilDry, CropWheat.

(b) Negate goal: ¬ApplyDripMethod.
Resolve P1 + SoilDry → IrrigationNeeded. Resolve with P2 → ¬CropWheat□ApplyDripMethod.
Resolve with CropWheat → ApplyDripMethod. Resolve with ¬ApplyDripMethod → empty clause □.
ApplyDripMethod proven true.

(c) If SoilDry removed: IrrigationNeeded can no longer be derived, so ApplyDripMethod is not derivable (now false/unknown under closed-world assumption). Since ¬ApplyDripMethod□CropWheat holds, P3 still fires and CropAtRisk still follows - the diagnosis path changes but the conclusion is the same.

Case Study 4: Wumpus World Knowledge Agent

(a) Belief state via Modus Ponens: Stench[1,2] → Wumpus adjacent to [1,2] (candidates [1,1]/[2,2]/unexplored neighbor). Breeze[1,1] → Pit adjacent to [1,1] (candidates [1,2]/[2,1]). Glitter[2,2] → Gold(2,2). Derived facts: Gold(2,2); Wumpus-adjacent(1,2); Pit-adjacent(1,1).

(b) Forward Chaining: Start with percepts as facts. Fire Stench-rule → possible Wumpus near [1,2]. Fire Breeze-rule → possible Pit near [1,1]. Fire Glitter-rule → Gold at [2,2]. Since [2,2] shows Glitter with no Stench/Breeze reported there, [2,2] is inferred hazard-free, so the Wumpus/Pit are placed at the remaining unexplored neighbor cells, not at [1,1] or [2,2].

(c) Path [1,1]→[1,2]→[2,2]: [1,2] itself has no Breeze percept, so no Pit there (Stench indicates Wumpus nearby, not necessarily in that cell). [2,2] has Glitter but no Stench/Breeze, so by ¬Wumpus□¬Pit→Safe, Safe(2,2) holds. Conclusion: the path is safe and the agent can proceed to collect the gold.